

Socioeconomic and Behavioral Determinants of Oral Health Outcomes Based on NHANES Data (2013–2018)

Authors: Rithwik Raj Suram, Vaishnavi Madhiraju, Varshini Guthi

Faculty Advisor: Dr. Divya S. Subramaniam

Department of Health Data Science, Saint Louis University

Background

- Oral health is a key aspect of overall well-being, yet disparities persist among U.S. adults.
- Lower income, limited education, and smoking are consistently associated with untreated dental caries and tooth loss, reflecting broader patterns of social disadvantage and unequal access to preventive care.
- Socioeconomic and behavioral factors particularly **income, education, and smoking** play a major role in determining oral health outcomes (Griffin et al., 2012; Dye et al., 2015).
- Understanding these relationships informs equitable interventions that expand preventive care and reduce disease burden

Objective

To analyze how income, education, and smoking status influence oral health outcomes using data from **NHANES 2013–2018**, and to generate insights supporting evidence-based policy actions for improving oral-health equity.

Methods

Data Source

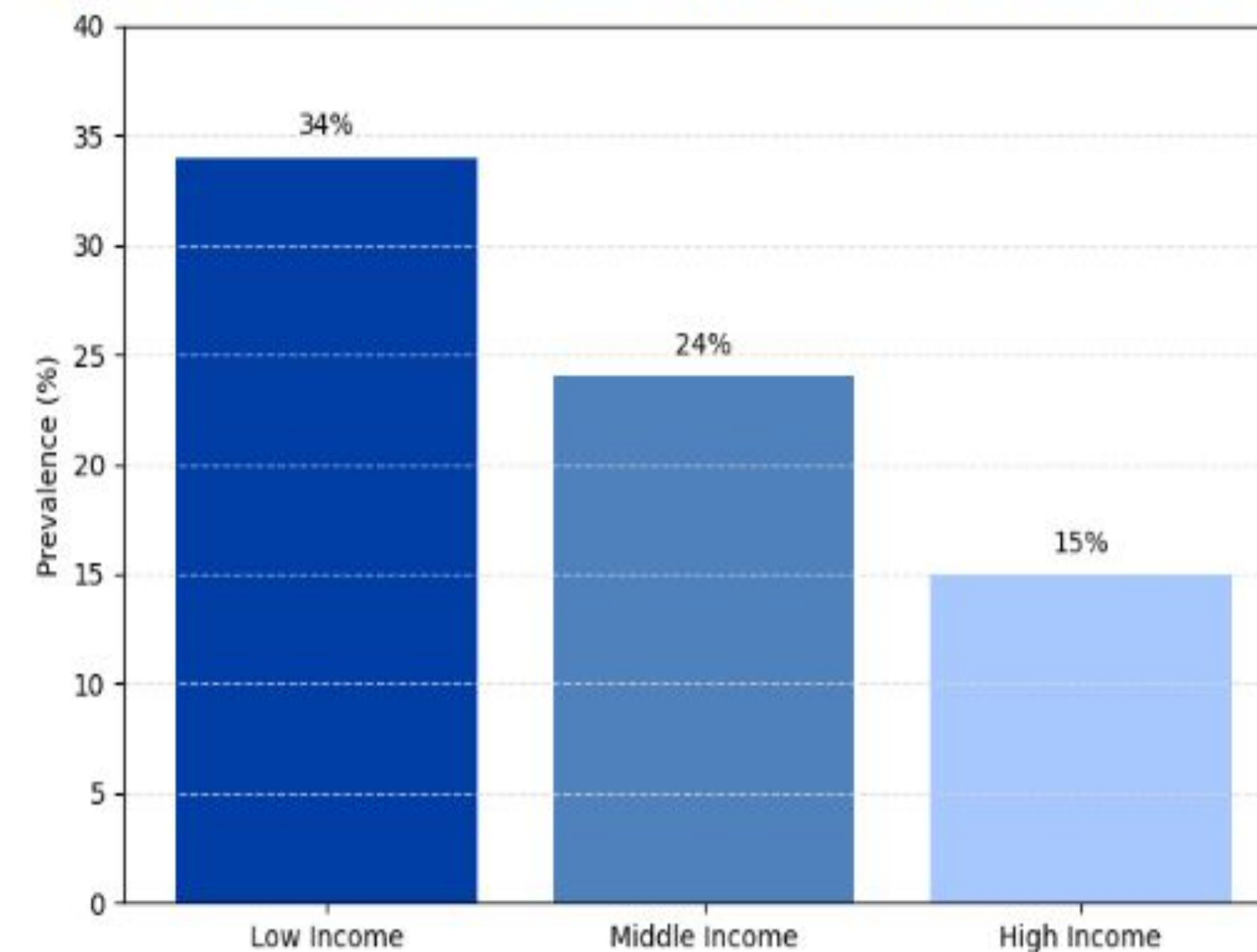
National Health and Nutrition Examination Survey (NHANES) 2013–2018 (~25,000 adults aged ≥ 18). Three survey cycles (2013–2014, 2015–2016, 2017–2018) were merged using participant ID (SEQN).

Variables:

- Outcomes:** Untreated dental caries (binary), missing teeth (count)
- Predictors:** Income-to-poverty ratio, education, smoking status
- Confounders:** Age, sex, race/ethnicity, dental insurance
- Statistical Analysis**

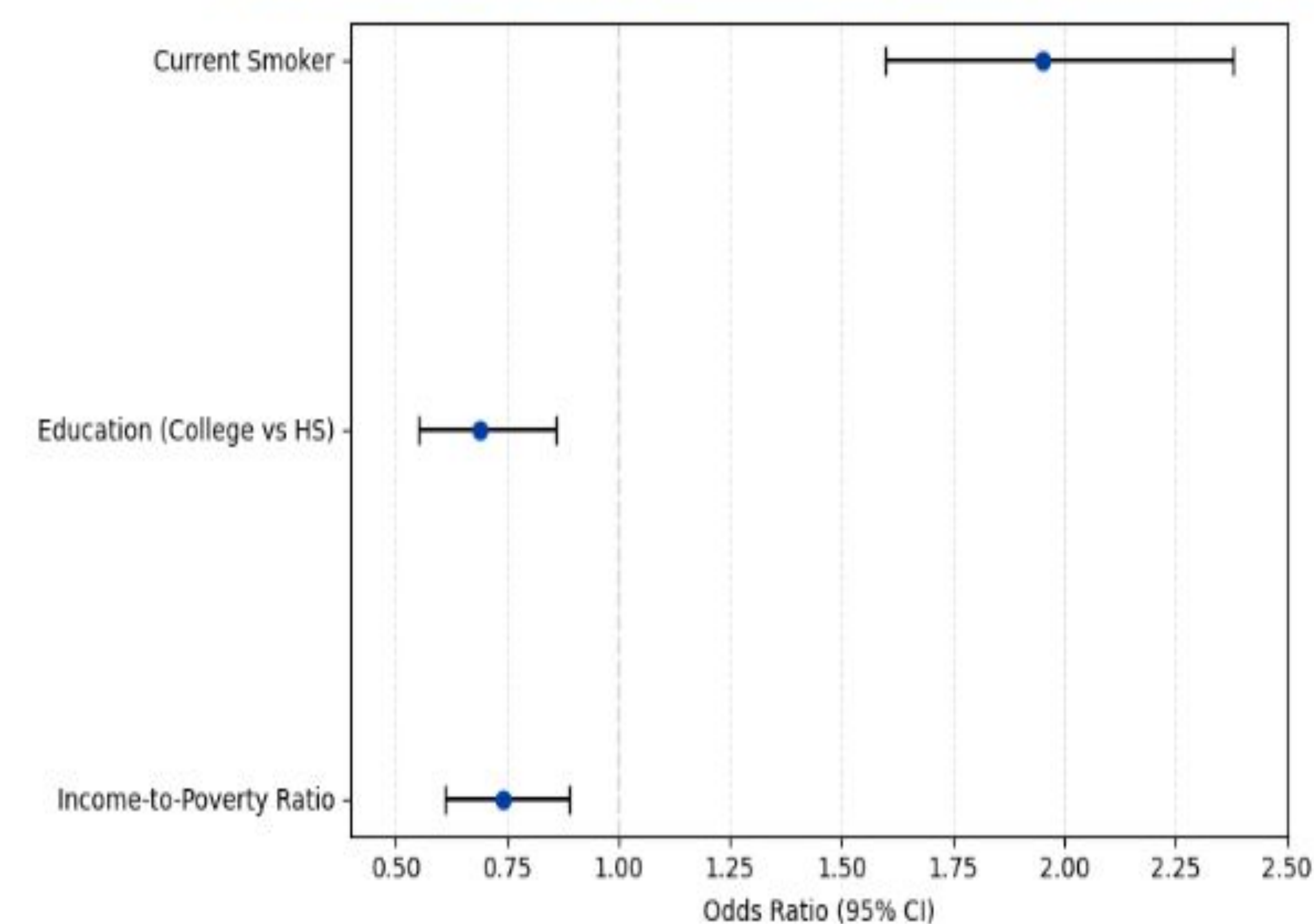
Data were cleaned, harmonized, and weighted per NCHS (2018) guidelines. Descriptive and bivariate analyses (χ^2 , t-tests) were conducted. Logistic regression modeled untreated caries, and linear regression modeled missing teeth. Sensitivity analyses were stratified by age and insurance coverage. Missing data were handled using listwise deletion.

Figure 1. Prevalence of Untreated Dental Caries by Income Level (NHANES 2013–2018)



Caries prevalence inversely correlated with income (Low: 34%, Middle: 24%, High: 15%).

Figure 2. Adjusted Odds Ratios for Socioeconomic and Behavioral Predictors of Poor Oral Health



Smoking and poverty strongest predictors; dashed line at OR = 1 indicates no effect.

Results

Descriptive Findings:

- Untreated dental caries: 34% among low-income adults vs. 15% among high-income.
- Missing teeth counts higher among less educated and current smokers.
- Lack of dental insurance associated with more missing teeth.

Regression Findings:

Predictor	Odds Ratio (95% CI)	p-value
Income-to-poverty ratio ↑	0.74 (0.61–0.89)	0.002
Education (College vs HS)	0.69 (0.55–0.86)	0.004
Current smoker	1.95 (1.60–2.38)	< 0.001

Model Fit: $R^2 \approx 0.28$ (moderate explanatory power).

Interpretation: Lower income, limited education, and smoking nearly double the odds of poor oral health outcomes.

Conclusion

- Socioeconomic and behavioral determinants strongly predict oral-health disparities among U.S. adults.
- Individuals with lower income, limited education, and smoking habits experience significantly higher risks of untreated dental caries and tooth loss.
- Insurance mitigates but does not eliminate disparities.

Findings align with national literature (Sabbah et al., 2007; Williams et al., 2008).

Public Health Implications:

Expanding preventive dental coverage, improving oral-health literacy, and integrating smoking cessation into community programs can reduce disparities and promote health equity.

Key Takeaway:

Income, education, and smoking collectively explain ~28% of the variance in oral health outcomes.